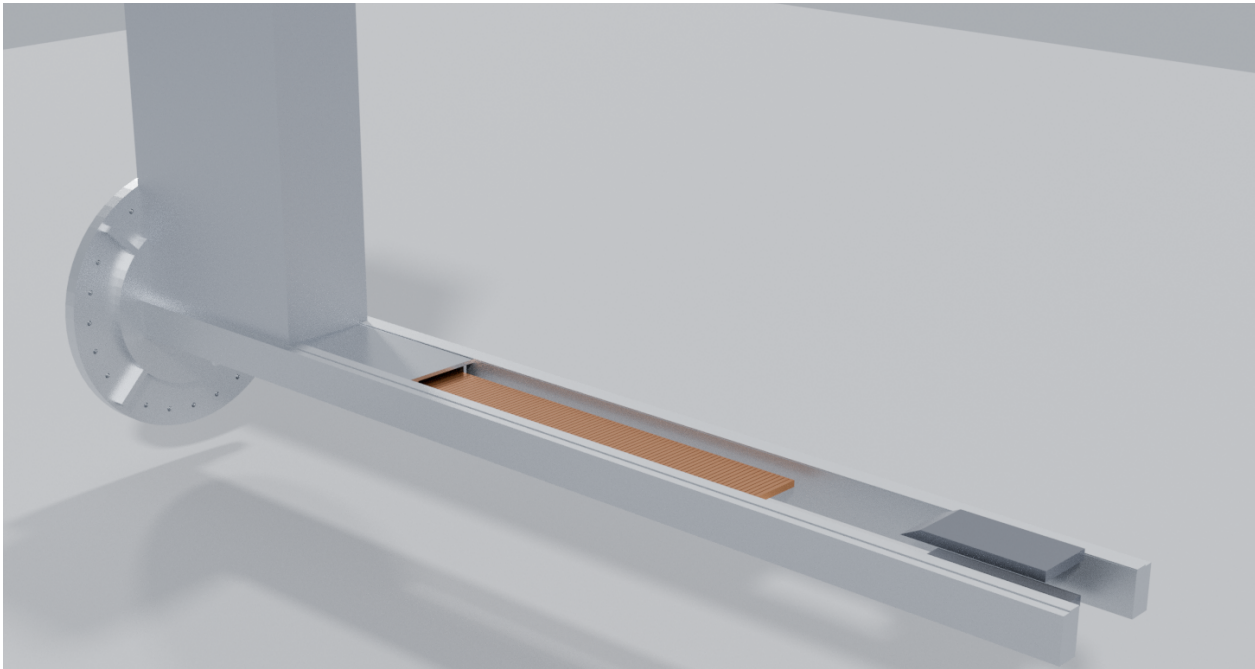


VOLLEY

An engineering study of **commanded, per-satellite orbit change** for CubeSats that are never modified.

Adityavardhan Mishra
B.Tech Mechanical Engineering
Symbiosis Institute of Technology
Symbiosis International (Deemed University), Pune
adityavardhanmishra@gmail.com

A CubeSat flown as a rideshare secondary inherits the primary customer's orbit. The spring that ejects it adds 1–2 m/s — enough to drift clear of the stage, not enough to change an orbit — and of the more than **4 800 nanosatellites and CubeSats catalogued** as of January 2026, on the order of **222 carry a propulsion system**, so the rest stay where they were dropped for life. VOLLEY asks what it costs to give each of them a *commanded* departure velocity without putting anything on the satellite.



Gen5 analysed architecture — CAD model, not flight hardware. Ironless double-sided Halbach linear synchronous motor, 1.5 m track, reusable magnet sled, twelve 3U satellites from two transverse cassettes, contactless eddy-current arrest. Nothing shown here has been built, fired or measured.

THE PROGRAMME, AND WHAT EACH LAYER IS WORTH

VOLLEY	the programme	one question, asked to a fixed evidence standard: can a propulsion-less satellite be given its own orbit?
Gen5	the analysed evidence base	a self-contained electromagnetic deployer. 65 run sheets, four independent solvers. Everything numeric on this sheet is Gen5.
Current direction	stage-integrated, gas-driven	the payload accelerated directly along a rail a spent upper stage already provides. It has not inherited Gen5's validation and nothing here is claimed for it.

THE CONCEPT OF OPERATIONS



The host is an active participant, not a mounting surface. A 100 m/s host budget reaches four shells and a fleet spanning 269 km of altitude; plane change is excluded at 133 m/s per degree. The altitude half of any multi-orbit claim is bought with host propellant, not with this machine.

GEN5, AT THE RATED POINT

Thrust constant ripple	10.54 N per kA/m ±1.01 % to 0.03 % to 0.059 %	Exit velocity, 3U acceleration	16.029 m/s 10.07 g of a 25 g cap
2-D FEM agrees		Dispersion, closed loop as apogee	0.0274 m/s (3σ) ±0.10 km
3-D FEM agrees		Semi-major axis change	+28.8 km
Energy per shot net of recovery to the payload	2782 J gross 2735 J 514 J — 18.8 %	Orbital lifetime against a spring	× 1.60 at mean activity +60.2 % vs +8.2 %
Pulse	162.3 ms, 320 A peak	Mass, dry / loaded per 3U satellite	126.6 / 174.6 kg
Track first mode	109.0 Hz	Recurring cost	10.55 kg ₹ 1,345,055 — every price assumed
Recoil per shot	64.1 N·s		

Every figure above is a model output, single-sourced to a committed JSON result and re-checked against its script on every commit. **None of it is a measurement, and no vendor has quoted any line of the cost.**

The contribution is the combination, not any one part of it. Electromagnetic launch of CubeSats has been studied before, at velocities twenty times higher — but at accelerations of order $10^3 g$ and with a conductive armature bonded to the payload. What is claimed here is **a commanded per-satellite velocity, on a satellite that is never modified, at an acceleration inside its existing qualification envelope, from a stage that was going to be discarded.** Each of those alone is ordinary. Together they describe a service tier with no demonstrated occupant.

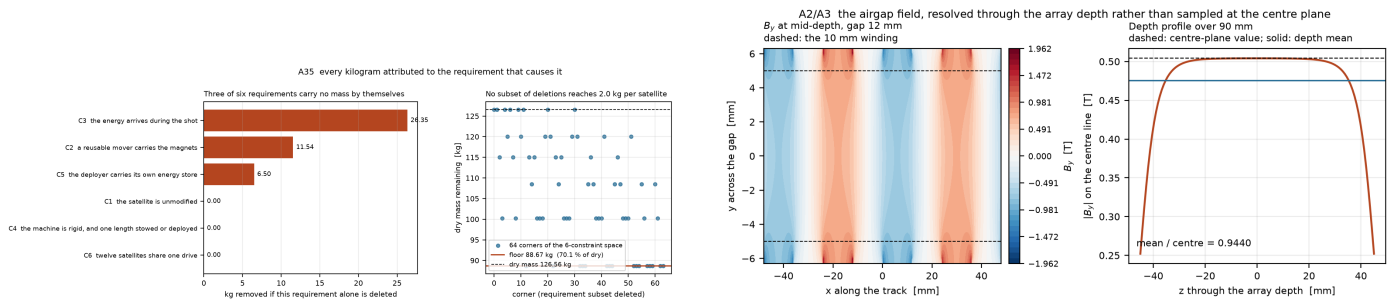
How the numbers were made, and what they cost the argument

THE METHOD CHAIN

Geometry	Parametric CAD built from one parameter file; 23 dimensions read back out of the model and checked against it.
Electromagnetics	Analytic travelling-wave model, cross-checked by magpylib , then by a 2-D magnetostatic FEM written from the weak form (scikit-fem, 141 k elements) to 0.03 %, then by a 3-D reduced-scalar-potential solve (getdp, 274 105 DoF) to 0.059 %.
Structures	CalculiX on the sled chassis and the retention chain.
Aerodynamics	OpenFOAM simpleFoam external solve, 581 779 cells – for the air correction a ground test would need, not for flight.
Circuit	ngspice transient on the pulse chain, against the analytic ledger.
Astrodynamics	GMAT R2022a headless, MSISE90, 20×20 gravity, lunisolar and SRP, against this project’s own propagator.
Control	Closed-loop Monte Carlo, 800 runs, with modelled sensing.
The discipline	65 run sheets. Each declares its acceptance bands, and commits them, before the script that produces the number exists – provable by checking that the band commit touched no script. A failed band moves the design point. No band has ever been widened after a result was known.

WHAT THE ANALYSIS CHANGED – THE FOUR THAT COST THE MOST

- 1
- Exact CAD mass took the headline from 20.37 to 16.029 m/s. A parametric 4.86 kg sled became a 9.445 kg one when solid volumes were computed; resolving the field through the array’s 90 mm depth, rather than sampling the centre plane, then took another 4.42 %. *Both consequences were declared in writing before the analyses ran.*
- 2
- An independent propagator falsified a claim in this project’s own abstract. The orbital-lifetime ratio had been offered as invariant across ballistic coefficient and solar activity. GMAT disagreed at low activity by 18 % against a 5 % band. The cause was the sweep, not the arithmetic: both variables entered the same multiplicative slot, so the ratio could not have moved. *The claim is withdrawn in the manuscript.*
- 3
- ngspice found a term no script carried. Energy closure failed at 97.0 % against a declared 98–102 %. The missing **86.6 J a shot** was the capacitor bank’s own series resistance, which the analytic ledger had no term for at all.
- 4
- Every kilogram was attributed to the requirement causing it, and then every requirement was deleted. Across all **64 corners**, **88.67 kg survives** – 70.06 % of dry mass, **7.39 kg per satellite against a 2 kg criterion**. *This is why Gen5 cannot be optimised into its own mass target, and it is the single result that redirected the work.*



Left: the constraint ledger – what each requirement costs alone, and the floor no combination of deletions reaches. Right: the airgap field, and the depth profile that made the thrust constant 4.42 % smaller. Both are model outputs.

THE HONEST COMPARISON

	Spring dispenser	VOLLEY Gen5	Cold-gas module on the satellite
Δv per satellite	1–2.5 m/s, one value for all	16.03 m/s, commanded	~16 m/s
Designed differential	exactly 0	16.03 m/s, to 0.027 (3 σ)	per satellite
Orbital energy	negligible	+28.8 km, ×1.60 life	comparable
Mass per 3U sat.	~6.0 kg	10.55 kg – loses, 1.758 ×	0.5–1.2 kg – loses, 12.4 ×
Satellite carries	nothing	nothing	pressure vessel, qualification, range safety
Host provides	one deploy signal	150–300 W, serial link, firing window	nothing
Maturity	TRL 9, thousands flown	TRL 2–3, nothing measured – loses	COTS, flown

Two of the three rows that matter commercially are losses, and they are printed because they are the result. Mass parity with a canister of springs was claimed by this project and is **withdrawn**: an acceptance band asked for parity within 15 % and the measured ratio is 1.758. So is the constellation-phasing claim – 30° of in-track phase costs **468 seconds of waiting** at zero Δv , so a spring and a clock reach it and this machine is not needed for it. **What survives is the third row: only Δv changes an orbit, and a clock cannot.** No cost comparison is offered in either direction; there is no quotation for any of it.

WHERE IT GOES NEXT



The next step is not more analysis. It is a benchtop track and a mass simulator – the point at which the *measured* evidence class in this project’s own figure index stops having zero members.

Status. Phase I design study, TRL 2–3. Nothing has been built, fired, measured, qualified or flown, and no result here has been reviewed by a third party. Model-to-model agreement between two solvers is a consistency check, not experimental validation. The defect register carries **132 numbered entries, 53 of them still open**, including every one that damages the claims above. **No institutional, agency or commercial affiliation is claimed or implied by this document.**



The full record
github.com/
aaaaaaaaaavm/VOLLEY